

CE/CZ4055: CPS Security
Tutorial 2

- (1)(a) An Encryption scheme comprises of a set of deterministic algorithms $(\mathcal{E}, \mathcal{D})$ and is defined over the message space $\mathcal{M}$, ciphertext space $\mathcal{C}$ and key space $\mathcal{K}$ as follows:

Encryption: $c = \mathcal{E}(k, m) = k \oplus m \mid c \in \mathcal{C}, k \in \mathcal{K}, m \in \mathcal{M}$

Decryption: $m = \mathcal{D}(k, c) = c \oplus k \mid c \in \mathcal{C}, k \in \mathcal{K}, m \in \mathcal{M}$

Alice wants to send messages in English to Bob, but decides to encrypt the messages using the above mentioned encryption scheme. She encrypts n messages $(m_1, m_2, \dots, m_n)$ with $n/2$ keys $(k_1, k_2, \dots, k_{n/2-1})$ already shared with Bob such that k_i is used to encrypt messages (m_{2*i}, m_{2*i+1}) for $i \in \{1, 2, \dots, n\}$. Eve passively eavesdrops on this communication from Alice to Bob and could only retrieve the encrypted ciphertexts sent by Alice. Can Eve retrieve the messages from the encrypted ciphertexts. If the encryption scheme is secure, please provide formal proof of security for the same. If not, please provide a possible attack that can be mounted by Eve to retrieve the encrypted messages.

- (b) Alice and Bob use secure One Time Pad ciphers for their two way communication. They use same set of n keys $(k_1, k_2, \dots, k_n)$ to sign a set of n messages $(m_1, m_2, \dots, m_n)$. Is their communication secure?

- (c) The One Time Pad cipher is a perfectly secure cipher. But, One Time Pads are rarely used for practical applications. Please discuss the reasons for the same.

- (2) Assume a Pseudo Random Number Generator (PRNG) $G : K \leftarrow \{0, 1\}^* \mid K \in \{0, 1\}^n$ outputs a random sequence when given an input seed K of length n such that $\text{XOR}(G(K)) = 1$. Is the Pseudo Random Number Generator really pseudo random? If not, state reasons why it is not.

- (3) (The 802.11b insecure MAC). Consider the following MAC (a variant of this was used for WiFi encryption in 802.11b WEP). Let F be a PRF defined as $F : \{0, 1\}^{128} \times \{0, 1\}^{128} \rightarrow \{0, 1\}^{32}$. Let CRC32 be a simple and popular error-detecting code meant to detect random errors; $\text{CRC32}(m)$ takes inputs $m \in \{0, 1\}^*$ and always outputs a 32-bit string. For this exercise, the only fact you need to know is that $\text{CRC32}(m_1) \oplus \text{CRC32}(m_2) = \text{CRC32}(m_1 \oplus m_2)$. Define the

following MAC system ($Sign, Verify$) as:

$Sign(k, m) = r \leftarrow \{0, 1\}^{128}, t = F(k, r) \oplus CRC32(m)$, Output: (r, t)

$Verify(k, m, (r, t)) = \{\text{accept if } t = F(k, r) \oplus CRC32(m) \text{ or reject otherwise}\}$

Show that this MAC system is insecure.